5. **In Hartley oscillator high frequency currents are prevented from using the power supply as a return path by**
(a) insertion of low reactance radio frequency choke in series
(b) insertion of a high reactance radio frequency choke in series
(c) insertion of a high resistance in series
(d) insertion of a capacitor in series

WBPSC AE 2003

Ans. (b) :

6. **Internal device capacitances become more prominent in**
(a) DC operation
(b) High frequency region
(c) Low frequency region
(d) Ac operation

UPRVUNL AE 2014

Ans. (b) Internal device capacitance become more prominent in high frequency region.

7. **Hartley oscillator generates**
(a) Very low frequency oscillation
(b) Radio-frequency oscillation
(c) Microwave oscillation
(d) Audio-frequency oscillation

UPRVUNL AE 2014

Ans. (b) Hartley oscillator is a type of harmonic oscillator. These are the tuned circuit oscillators which are used to produce the waves in the range of radio frequency and hence are also referred to as radio frequency oscillator.
Frequency of oscillation of Hartley oscillator is:-

$$\omega = \frac{1}{\sqrt{LC}}$$

Where, $\left[L = L_1 + L_2 \pm 2M\right]$

8. **The current gain A_I in a small signal transistor amplifier is-**

(a) $A_I = \frac{-h_f}{1+h_0Z_L}$ (b) $A_I = \frac{h_f}{1+h_0Z_L}$
(c) $A_I = \frac{-h_i}{1+h_0Z_L}$ (d) $A_I = \frac{h_i}{1+h_0Z_L}$

UPPSC AE 2011, Paper-II

Ans. (a) : Current Gain $(A_i) = \frac{\text{Output current}}{\text{Input current}}$

$$A_i = \frac{I_L}{I_1} = \frac{-I_2}{I_1}$$

$$I_2 = h_fI_1 + h_0V_2 \quad ......(1)$$

$$V_2 = -I_2Z_L$$

Put the value of V_2 in equation (1)

$$I_2 = h_fI_1 - h_0I_2Z_L$$

$$I_2 + I_2Z_Lh_o = h_fI_1$$

$$I_2(1 + h_0Z_L) = h_fI_1$$

$$\boxed{\frac{I_2}{I_1} = \frac{h_f}{1+h_0Z_L}}$$

$$\boxed{A_i = \frac{-I_2}{I_1} = \frac{-h_f}{1+Z_Lh_o}}$$

9. **For a wide range of oscillation in the audio range the preferred oscillator is**
(a) Hartley (b) Phase shift
(c) Wien bridge (d) Hartley and Colpitt

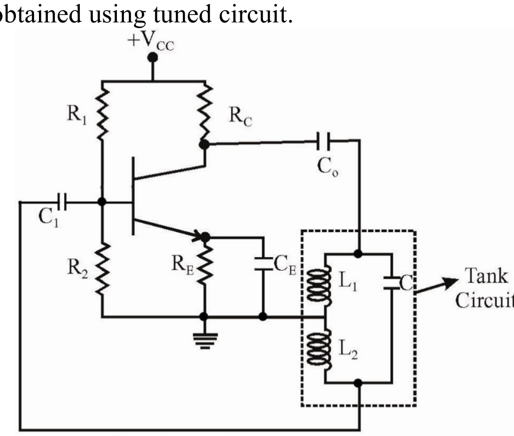
WBPSC AE 2007

Ans. (c) :

10. **The oscillator which uses a tapped coil in the LC circuit is known as**
(a) Colpitts Oscillator (b) Hartley Oscillator
(c) Armstrong Oscillator (d) Pierce Oscillator

MPPSC AE 2016

Ans. (b) : In Hartley oscillator the resonance frequency is obtained using tuned circuit.



Hartley oscillator

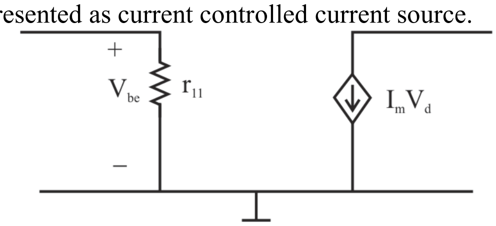
$$\boxed{f = \frac{1}{2\pi\sqrt{L_{eff}C}}}$$

Where $L_{eff} = L_1 + L_2$ ..... For different core
$L_{eff} = L_1 + L_2 + 2M$ .... for single core

11. **Which one of the following statement is true?**
(a) Transistor can be modelled as current controlled current source
(b) Transistor can be modelled as current controlled voltage source
(c) Transistor can be modelled as voltage controlled voltage source
(d) Transistor can be modelled as voltage controlled current source

MPPSC AE 2016

Ans. (a) : When we make ac model of transistor it can be represented as current controlled current source.



12. Frequency of oscillation of Colpitt's oscillator is

(a) $f = \dfrac{1}{2\pi\sqrt{L\left(\dfrac{C_1C_2}{C_1+C_2}\right)}}$

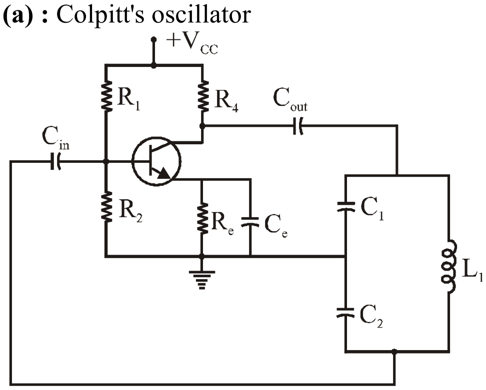
(b) $f = \dfrac{1}{2\pi\sqrt{L(C_1+C_2)}}$

(c) $f = \dfrac{1}{2\pi\sqrt{(L_1+L_2)C}}$

(d) $f = \dfrac{1}{2\pi\sqrt{(L_1+L_2+2M)C}}$

TSTRANSCO AE 2018

Ans. (a) : Colpitt's oscillator



Frequency of oscillation

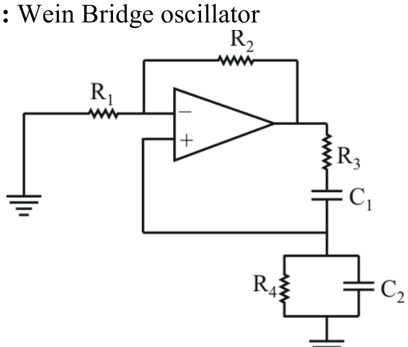
$$f = \frac{1}{2\pi\sqrt{L\left(\frac{C_1C_2}{C_1+C_2}\right)}}$$

13. The type of feedback used in wien bridge oscillator is

(a) Negative feedback only
(b) Positive feedback only
(c) No feedback
(d) Both negative and positive feedback

TSTRANSCO AE 2018

Ans. (d) : Wein Bridge oscillator



- A Wien bridge oscillator is a type of electronic oscillator that generates sine waves. It can generate radio frequencies.
- A Wien-bridge oscillator has two paths for feedback. It uses both positive and negative feedback with one path each.

14. Frequency of oscillation of Colpitts oscillator is given by :-

(a) $f_0 = \dfrac{1}{2\pi\sqrt{L/C_{eq}}}$ (b) $f_0 = \dfrac{1}{2\pi\sqrt{LC_{eq}}}$

(c) $f_0 = \dfrac{1}{2\pi\sqrt{C_{eq}/L}}$ (d) $f_0 = \dfrac{1}{2\pi}\sqrt{\dfrac{L}{C_{eq}}}$

UKPSC AE 2013 Paper-I

Ans. (b) : Frequency of oscillation of colpitts oscillator

$$f_0 = \frac{1}{2\pi\sqrt{LC_{eq}}}$$

Where, $C_{eq} = C_1C_2/C_1 + C_2$

Colpitts oscillator is one of the type of LC oscillator which consisting of resonant tank circuit. The tank circuit of two series capacitor parallel with a inductor.

15. The highest frequency stability is achieved by using an oscillator of the type:

(a) Colpitts (b) Crystal controlled
(c) Hartley (d) RC oscillator

Punjab PSC SDE 2017

Ans. (b) : The use of electric crystals in parallel resonant circuit provide high frequency stability in oscillators such oscillators are called as crystal oscillators.

- They have a high order of frequency stability.
- The quality factor (Q) of the crystal is very high.

16. Select from the options, the percentage duty cycle of the output voltage waveform for an astable multivibrator if R_A= 1KΩ and R_B = 2KΩ.

(a) 70% (b) 55%
(c) 50% (d) 60%

DMRC AM 2018

Ans. (d) : Given that, $R_A = 1\ k\Omega$, $R_B = 2\ k\Omega$

As we know that,

$$D = \frac{R_A + R_B}{R_A + 2R_B}$$

$$= \frac{1k + 2k}{1k + 2 \times 2k} = \frac{3}{5} = 0.6$$

So, 60% duty cycle.

17. For a sine wave oscillator, total phase shift around the loop and magnitude of loop gain must be-

(a) zero and unity
(b) 180^0 and unity
(c) zero and less than unity
(d) 90^0 and unity

KPSC ASTT. INSP. 2014

Ans. (a) : For a sine wave oscillator, total phase shift around the loop and magnitude of loop gain must be zero and unity.